

A fitness-independent simultaneous amplifier beyond fitness $3/2$

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Abstract

We study fixation from a uniformly random single mutant on finite connected loopless undirected weighted graphs under both Birth–death (Bd) and death–Birth (dB) Moran updating. Let R_{sim} be the supremum of the fitness intervals $(1, R)$ on which one graph family, chosen independently of fitness, eventually amplifies fixation under both rules. Let R_{hyb} be the unique root in $(3/2, 151/100)$ of

$$R^6 - 8R^5 + 22R^4 - 30R^3 + 21R^2 - 6R + 1.$$

We construct one fitness-independent family that simultaneously amplifies every fixed $1 < r < R_{\text{hyb}}$ for all sufficiently large members of the sequence. Therefore $R_{\text{sim}} \geq R_{\text{hyb}} = 1.5028569127905696 \dots > 3/2$.

The construction combines a large clique with dilute hub pendants and dilute, internally heavy two-vertex satellites. An exact finite weak-cut Schur trace, together with uniform establishment, cleanup, and reciprocal-invasion estimates at the scale of the eventual gain, reduces the two normalized fixation corrections to explicit response functions. Optimizing their common positivity interval yields the displayed sextic. The same analysis proves that R_{hyb} is optimal within this pair-pendant leading-order architecture; a fully rational specialization already crosses $3/2$. Thus $3/2$ is not the exact simultaneous-amplification threshold. The unrestricted value of R_{sim} , and even a finite universal upper bound, remain open.

Keywords: evolutionary graph theory; Moran process; fixation probability; simultaneous amplification; Birth–death updating; death–Birth updating.

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1 Introduction

Evolutionary graph theory extends the finite-population Moran process by representing sites as vertices and replacement opportunities as weighted edges (Moran, 1958; Lieberman et al., 2005; Allen et al., 2017). The order of the two elementary Moran events matters: stars and many other structures that amplify selection when birth precedes death suppress it when death precedes birth (Hindersin and Traulsen, 2015; Allen et al., 2020). Random edge weights also tend to make undirected networks less amplifying, and often suppressing, which emphasizes that useful weighting is a design problem rather than a generic effect (Bhaumik and Masuda, 2024). Death–Birth amplification is necessarily transient in fitness on every fixed noncomplete strongly connected structure (Tkadlec et al., 2020), and transient dB amplifiers have also been sought through guided spectral

edge deletion (Richter, 2023). A distinct approach randomizes the update order within a single mixed Moran process (Brewster et al., 2026). Here robustness instead means that the same graph beats the well-mixed benchmark under each pure rule separately. These tensions make robustness to update order a natural design problem rather than an automatic consequence of ordinary amplification.

A population structure is a *simultaneous amplifier* at fitness r when it raises uniformly initialized fixation probability, relative to the complete graph of the same order, under both Bd and dB updating. The asymptotic problem has the quantifier order

$$\exists\{G_t\} \forall r \in (1, R) \exists t_0(r) \forall t \geq t_0(r).$$

The graphs and weights may depend on t , but not on r .

Svoboda et al. (2024) constructed simultaneous amplifiers on $1 < r < 1.2$. A preceding public manuscript established $R_{\text{sim}} \geq 3/2$ with a center-triangle family, while that particular family suppresses at its endpoint (Kriebel, 2026). The present construction crosses $3/2$. The numerical increment is small, but the conclusion is qualitative: the earlier endpoint is not the unrestricted phase boundary.

The proof has four ingredients. First, a finite Schur complement gives the rare-migration trace without an independence or branching approximation. Second, stopped birth-death comparisons control establishment and the entire path to fixation at the smaller q/C scale of the final gain. Third, the exact two-coordinate macro chain retains adverse reversals while deriving the pair gate. Finally, optimizing the two resulting response functions produces an algebraic tangency and the degree-six polynomial in the abstract. This also proves optimality within the stated pair-pendant leading-order architecture. No universal upper bound is inferred; the unrestricted value of R_{sim} remains open.

2 Model, baselines, and threshold

Let $G = (V, w)$ be finite, connected, loopless, undirected, and weighted:

$$w_{uv} = w_{vu} \geq 0, \quad w_{uu} = 0, \quad d_u := \sum_v w_{uv} > 0.$$

A state is a mutant set $S \subseteq V$. Mutants have fitness $r > 1$ and residents fitness one. Under Bd, a parent is selected proportional to fitness and then a replacement target proportional to edge weight. Under dB, a uniformly chosen target dies and then a parent is selected proportional to fitness times edge weight. Thus, with $F(S) = |V| + (r - 1)|S|$,

$$P_{\text{Bd}}(S, S \cup \{v\}) = \frac{r}{F(S)} \sum_{u \in S} \frac{w_{uv}}{d_u}, \quad v \notin S, \quad (1)$$

$$P_{\text{Bd}}(S, S \setminus \{v\}) = \frac{1}{F(S)} \sum_{u \notin S} \frac{w_{uv}}{d_u}, \quad v \in S, \quad (2)$$

$$P_{\text{dB}}(S, S \cup \{v\}) = \frac{1}{|V|} \frac{r \sum_{u \in S} w_{uv}}{r \sum_{u \in S} w_{uv} + \sum_{u \notin S} w_{uv}}, \quad v \notin S, \quad (3)$$

$$P_{\text{dB}}(S, S \setminus \{v\}) = \frac{1}{|V|} \frac{\sum_{u \notin S} w_{uv}}{r \sum_{u \in S} w_{uv} + \sum_{u \notin S} w_{uv}}, \quad v \in S. \quad (4)$$

The fixation function h_U is the unique harmonic solution with boundary values zero at $\emptyset$ and one at V . We use

$$\rho_U(G, r) = \frac{1}{|V|} \sum_{v \in V} h_U(\{v\}), \quad U \in \{\text{Bd}, \text{dB}\}.$$

Direct solution of the one-dimensional complete-graph count chains gives

$$\rho_{\text{Bd}}(K_n, r) = \frac{1 - r^{-1}}{1 - r^{-n}}, \quad (5)$$

$$\rho_{\text{dB}}(K_n, r) = \frac{n-1}{n} \frac{1 - r^{-1}}{1 - r^{-(n-1)}}. \quad (6)$$

Both converge, for fixed $r > 1$, to

$$p(r) := 1 - \frac{1}{r}.$$

Definition 1. R_{sim} is the supremum of all $R > 1$ for which one fitness-independent family $\{G_t\}$, with $|V(G_t)| \rightarrow \infty$, satisfies

$$\rho_{\text{Bd}}(G_t, r) > \rho_{\text{Bd}}(K_{|V(G_t)|}, r), \quad \rho_{\text{dB}}(G_t, r) > \rho_{\text{dB}}(K_{|V(G_t)|}, r)$$

for every fixed $r \in (1, R)$ and every sufficiently large t .

3 The pair-pendant construction and main theorem

Let

$$P(R) = R^6 - 8R^5 + 22R^4 - 30R^3 + 21R^2 - 6R + 1$$

and let R_{hyb} be its unique root in $(3/2, 151/100)$. Define

$$\sigma_* = \frac{-R_{\text{hyb}}^3 + 4R_{\text{hyb}}^2 - 3R_{\text{hyb}} - 1}{2(R_{\text{hyb}} - 1)}, \quad (7)$$

$$\lambda_* = \frac{2(R_{\text{hyb}} - 1)(1 - \sigma_*)}{1 + \sigma_*(R_{\text{hyb}}^2 - 1)}. \quad (8)$$

Exact root isolation gives

$$R_{\text{hyb}} = 1.5028569127905696\dots, \quad \sigma_* = 0.1306772822870484\dots, \quad \lambda_* = 0.7508064830318805\dots$$

For $t \geq 2$, put

$$q_t = t, \quad m_t = \lfloor \lambda_* t \rfloor, \quad C_t = t^4, \quad n_t = C_t + m_t + 2q_t.$$

The graph contains:

1. a unit-weight clique on C_t vertices, one designated as the hub;
2. m_t unit-weight leaves adjacent only to the hub;
3. q_t disjoint two-vertex satellites, each with internal edge weight $W_t = C_t/\sigma_*$;
4. an edge of common weight $\varepsilon_t > 0$ from every satellite vertex to every clique vertex.

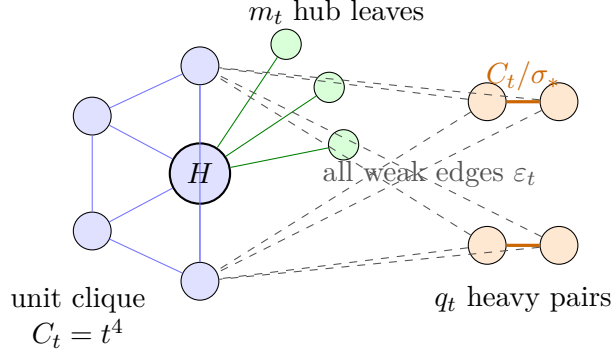


Figure 1: Schematic of the dilute hybrid. Dashed lines represent all satellite–clique edges, not only those drawn.

There are no other edges. In particular, there are no edges between distinct satellites.

To define the positive coupling without reference to a subsequently chosen fitness, let

$$I_t = [1 + 1/t, R_{\text{hyb}} - 1/t]$$

when this interval is nonempty. Let $\rho_{U,t}^0$ denote the finite separated trace defined in Section 4. Choose e_t to be the least positive integer such that, for $\varepsilon_t = 2^{-e_t}$,

$$\sup_{r \in I_t} \frac{n_t}{q_t} \left| \frac{\rho_U(G_t(\varepsilon_t), r)}{\rho_U(K_{n_t}, r)} - \frac{\rho_{U,t}^0(r)}{\rho_U(K_{n_t}, r)} \right| \leq \frac{1}{t} \quad (U = \text{Bd, dB}).$$

For the finitely many empty intervals, take $e_t = 1$. Finite Schur-complement continuity proves existence. At fixed t, e , all entries are algebraic rational functions of r , denominators are positive, and the two uniform inequalities are decidable by exact real-algebraic arithmetic. Thus this is an effective graph definition, and every weight is fixed before r is quantified.

Theorem 2 (Main theorem). *For every fixed $1 < r < R_{\text{hyb}}$, the family above satisfies, for every sufficiently large t ,*

$$\rho_{\text{Bd}}(G_t, r) > \rho_{\text{Bd}}(K_{n_t}, r), \quad \rho_{\text{dB}}(G_t, r) > \rho_{\text{dB}}(K_{n_t}, r).$$

Consequently

$$R_{\text{sim}} \geq R_{\text{hyb}} = 1.5028569127905696 \dots > 3/2.$$

4 Exact lumping and the finite weak-cut trace

A configuration has orbit label

$$(h, i, u, v, \ell),$$

where h is the hub type, i the number of mutant ordinary clique vertices, u the number of mixed pairs, v the number of all-mutant pairs, and ℓ the mutant-leaf count. The group

$$S_{C_t-1} \times (S_2 \wr S_{q_t}) \times S_{m_t}$$

is transitive on every fibre and commutes with both update kernels. This immediately gives the following exact reduction.

Lemma 3 (Strong orbit lumping). *For every positive value of the weak-edge weight and for both update rules, the total transition probability from a labelled configuration to a target fibre depends only on (h, i, u, v, ℓ) .*

Proof. Every permutation in the displayed wreath-product action preserves vertex degrees, edge weights, types, and fitnesses. It therefore commutes with each labelled parent–target event under Bd and each labelled death–parent event under dB. Summing events into a target fibre proves strong lumpability. $\square$

We next fix finite C, m, q and send $\varepsilon \downarrow 0$. At $\varepsilon = 0$ the closed fast classes are precisely the states in which the center module and every pair are internally homogeneous. Write a macro state as

$$(h, k) \in \{0, 1\} \times \{0, \dots, q\},$$

where h is the center type and k the number of mutant pairs. Partition the finite first-step matrix into mixed-module states F and all monomorphic-module states M , explicitly including global extinction and fixation. Their two harmonic values remain fixed at zero and one.

Proposition 4 (Finite weak-cut trace). *Let a_H^U, a_P^U be the uniformly initialized isolated fixation probabilities of the center and one pair, and let P_U^H, P_U^P be fixation probabilities of the limiting macro chain from $(1, 0)$ and $(0, 1)$. For $U \in \{\text{Bd}, \text{dB}\}$,*

$$\rho_U^0(C, m, q; r) = \frac{C + m}{N} a_H^U(r) P_U^H + \frac{2q}{N} a_P^U(r) P_U^P, \quad N = C + m + 2q. \quad (9)$$

Moreover, $\rho_U(G(C, m, q, \varepsilon), r) \rightarrow \rho_U^0(C, m, q; r)$ uniformly in r on every compact subinterval of $(0, \infty)$.

Proof. Order the states as M, F . At $\varepsilon = 0$, the first-step matrix has the block form

$$P_U(0) = \begin{pmatrix} I & 0 \\ C_0 & Q_0 \end{pmatrix}, \quad \rho(Q_0) < 1.$$

Indeed, from a mixed state each finite connected module absorbs internally almost surely, so $I - Q_0$ is invertible with nonnegative inverse and $(I - Q_0)^{-1}C_0$ is the stochastic matrix of its first monomorphic return. Expanding the positive-cut chain gives

$$P_{MM}(\varepsilon) = I + \varepsilon A + O(\varepsilon^2), \quad P_{MF}(\varepsilon) = \varepsilon B + O(\varepsilon^2),$$

and $P_{FM}(\varepsilon) = C_0 + O(\varepsilon)$, $P_{FF}(\varepsilon) = Q_0 + O(\varepsilon)$. Eliminating F from the finite harmonic system yields, after division by the common factor ε , the macro generator

$$\mathcal{L}_U = A + B(I - Q_0)^{-1}C_0. \quad (10)$$

The second term first makes one cross-module introduction and then resolves its entire zero-cut internal excursion. Its absorption distribution is therefore exactly the affected module's local fixation probability. On the two globally absorbing macro states we retain boundary values zero and one; on every other macro state, the leading harmonic equation is $\mathcal{L}_U h = 0$. Positive introduction rates make every interior state of this finite macro ladder transient. On a compact fitness interval its interior inverse is uniformly bounded, so the boundary-value problem has a unique solution. This is the claimed introduction-rate-times-fixation trace, derived from the finite chain rather than from independent lineages.

A uniformly chosen initial mutant first resolves inside its center or pair module with probability $1 - O(\varepsilon)$. Conditioning on which module fixes locally gives (9). All matrices are finite and their entries are continuous rational functions of (ε, r) with positive event denominators. After fixing the two boundary coordinates, the scaled Schur complement converges to the invertible interior restriction of $\mathcal{L}_U$. On a compact fitness interval that limiting inverse, $(I - Q_0)^{-1}$, and the nearby absorbing inverses are uniformly bounded. The block expansion and hence the convergence are uniform there. $\square$

5 Clique-pendant estimates at the gain scale

Let $H_{C,m}$ be a unit clique K_C with m unit hub pendants and put $p = 1 - 1/r$. Along $C = t^4$, $q = t$, and $m \asymp t$, the following estimates hold uniformly when r ranges over a compact subset of $(1, \infty)$:

Proposition 5 (Center module).

$$u_{\text{core}}^U(r) = p + o(q/C), \quad U = \text{Bd}, \text{dB}, \quad (11)$$

$$u_{\text{leaf}}^{\text{Bd}}(r) = 1 - o(1), \quad (12)$$

$$u_{\text{leaf}}^{\text{dB}}(r) = O(C^{-1}). \quad (13)$$

Here u_{core}^U averages the C clique starting vertices. Consequently

$$a_H^{\text{Bd}}(r) = \frac{Cp + m}{C + m} + o(q/C), \quad (14)$$

$$a_H^{\text{dB}}(r) = \frac{Cp}{C + m} + o(q/C). \quad (15)$$

At reciprocal fitness $1/r$,

$$u_{\text{core}}^{\text{Bd}}(1/r) = o(C^{-1}), \quad J_H(1/r) = o(C^{-1}),$$

where J_H is the dB portal functional defined below.

Proof. We give the quantitative argument because an $o(1)$ establishment approximation would be too weak: the final gain is only of order q/C . Write a center state as (h, i, ℓ) , put $c = C - 1$, $d = c + m$, and suppress a common positive statewise time factor. The six Bd changing rates are

change	rate
$i \rightarrow i + 1$	$r(c - i)\{h/d + i/c\}$
$i \rightarrow i - 1$	$i\{(1 - h)/d + (c - i)/c\}$
$0 \rightarrow 1$ at h	$r(i/c + \ell)$
$1 \rightarrow 0$ at h	$(c - i)/c + m - \ell$
$\ell \rightarrow \ell + 1$	$rh(m - \ell)/d$
$\ell \rightarrow \ell - 1$	$(1 - h)\ell/d$

After multiplication by $C + m$, the six dB rates are

$$\begin{array}{l|l}
i \rightarrow i+1 & (c-i) \frac{r(h+i)}{c + (r-1)(h+i)} \\
i \rightarrow i-1 & i \frac{c-i+1-h}{c + (r-1)(i-1+h)} \\
0 \rightarrow 1 \text{ at } h & \frac{r(i+\ell)}{d + (r-1)(i+\ell)} \\
1 \rightarrow 0 \text{ at } h & \frac{d-i-\ell}{d + (r-1)(i+\ell)} \\
\ell \rightarrow \ell+1 & h(m-\ell) \\
\ell \rightarrow \ell-1 & (1-h)\ell.
\end{array}$$

These tables follow by summing the labelled parent–target and death–parent events and are the only estimates used below.

Start from one ordinary clique mutant, with resident hub and no mutant leaves. Stop i on hitting 0 or $K = A \log C$. Before the first hub change, the exact embedded up/down odds are

$$R_{\text{Bd}}(i) = r \frac{c-i}{c-i+c/(c+m)}, \quad (16)$$

$$R_{\text{dB}}(i) = r \frac{c-i}{c-i+1} \frac{c+(r-1)(i-1)}{c+(r-1)i}. \quad (17)$$

Both equal $r\{1 + O(K/C)\}$. At $i \leq K$, the hub-change hazard divided by the ordinary-change hazard is $O(K/C)$. A biased walk started at one makes $O(K)$ expected ordinary changes before hitting $\{0, K\}$, uniformly on a fitness compact bounded away from one. Charging the first hub change as an error event therefore costs $O(K^2/C)$.

The product-odds gambler’s-ruin formula therefore gives

$$\mathbb{P}_1(T_K < T_0) = p + O(K^2/C) + O(r^{-K}).$$

For every hub and leaf state, the ordinary-count odds remain bounded below by a constant $r_0 > 1$ from K to a fixed positive core density. Indeed, the two tables give the uniform lower bounds

$$\frac{q_{\text{Bd}}^+}{q_{\text{Bd}}^-} \geq \frac{r}{1 + c/\{d(c-i)\}}, \quad \frac{q_{\text{dB}}^+}{q_{\text{dB}}^-} = r \frac{h+i}{i} \frac{c-i}{c-i+1-h} \frac{c+(r-1)(i-1+h)}{c+(r-1)(i+h)}.$$

Hence return to zero costs $O(r_0^{-K})$. Once i reaches a fixed positive density, the same comparison carries it to $i = (1-\delta)c$ with failure $\exp\{-\Omega(C)\}$. Write $R = c-i$ for the resident-core deficit. In the upper strip $1 \leq R \leq 2\delta c$, the two rate tables give, uniformly in the hub and leaf coordinates,

$$\frac{q(R \mapsto R-1)}{q(R \mapsto R+1)} \geq r(1-2\delta) \frac{R}{R+1} \{1 - o(1)\}. \quad (18)$$

Writing r_- for the lower end of the chosen fitness compact, choose

$$\delta < \min \left\{ \frac{1}{8}, \frac{r_- - 1}{8r_-}, \frac{r_-^2 - 1}{4(r_-^2 + 1)} \right\}.$$

Then one may take $\alpha = (r_- + 1)/2 > 1$ and, for all sufficiently large C , replace the right side of (18) by $\alpha R/(R+1)$. It is larger than $1 + \eta_0$ above the fixed integer $R_0 = \lceil (\alpha + 1)/(\alpha - 1) \rceil$;

no statewise drift assertion is made below R_0 . Exponential gambler's ruin between R_0 and $2\delta c$, followed by the strong Markov property at returns to R_0 , shows that for every fixed M , in the displayed continuous-time statewise clock,

$$\mathbb{P}\left(\sup_{s \leq C^M} R_s \geq 2\delta c \mid R(0) \leq \delta c\right) \leq C_M C^{M+1} e^{-\gamma C}. \quad (19)$$

The polynomial factor bounds the number of uniformized R -changes. The same comparison gives an $O(\log C)$ hitting time of the bounded strip and a geometric tail for each excursion away from it, except on the event in (19).

It remains to synchronize the hub and its pendants while the core is confined. Under Bd, freeze (i, R, ℓ) between hub and leaf events and put

$$A = r(i/c + \ell), \quad B = R/c + m - \ell, \quad u = \frac{r(m - \ell)}{c + m}, \quad v = \frac{\ell}{c + m}.$$

If p_h is the probability that the next leaf event is upward, conditional on the current hub type h , renewal at the next hub flip gives

$$p_0 = \frac{A}{A + v} p_1, \quad p_1 = \frac{u}{B + u} + \frac{B}{B + u} p_0.$$

Consequently the next-leaf up/down odds, from either hub phase, are bounded below by

$$\frac{Au}{Bv} \frac{1}{1 + u/B} = \frac{r^2(i/c + \ell)(m - \ell)}{(R/c + m - \ell)\ell} \{1 - O(C^{-1})\} \geq \frac{r_-^2}{1 + 2\delta} \{1 - o(1)\} > 1. \quad (20)$$

for $1 \leq \ell < m$. This lower bound survives core-count interruptions. Indeed, during an $h = 0$ phase the competing-hazard bound gives $\mathbb{P}(\text{leaf down before hub activation}) \leq v/(A + v)$; during an $h = 1$ phase it gives $\mathbb{P}(\text{leaf up before hub deactivation}) \geq u/(B + u)$. A core jump changes neither h nor ℓ and merely restarts one of these races with the same uniform bounds. The strong Markov property at successive hub cycles therefore gives eventual up/down odds at least the uniform final bound in (20). From $\ell = 0$, a mutant-leaf immigration occurs in $O(C)$ mean time. The leaf trace is therefore dominated below by a reflected biased walk and reaches m in $O(Cm)$ mean time. Once the leaves are mutant, a resident hub reactivates before the next leaf loss with probability $1 - O(C^{-1})$. While the hub is mutant, the ordinary deficit has inward odds bounded below by r_- and hub-loss hazard $O(R/C)$, so starting anywhere in the confined strip it has probability at least $e^{-O(\delta)} > 0$ of hitting $R = 0$ first. Thus one block of C^2 time succeeds with probability at least a fixed $s_0 > 0$. Repeating $O_B(\log C)$ blocks makes cleanup failure $O(C^{-B})$ for any prescribed fixed B .

Under dB, every pendant copies the hub at its next death. In the bounded R strip the hub activation rate is bounded below, whereas its deactivation rate when mutant is $O(m/C)$. An activated interval of length $B \log C$ therefore clears all pendants with failure at most mC^{-B} , drives R to zero with a compact-uniform subcritical tail, and is interrupted with probability $O((m/C) \log C) = o(1)$. Repeating $O_B(\log C)$ attempts again gives failure $O(C^{-B})$ in polynomial time. Combining either cleanup with (19) makes the probability of a resident core lineage returning to positive density $\exp\{-\Omega(C)\}$ throughout.

Choose A and then B uniformly on the fitness compact so that all of

$$O(K^2/C) + O(r^{-K}) + O(r_0^{-K}) + O(C^{-B}) + \exp\{-\Omega(C)\}$$

are $o(q/C) = o(C^{-3/4})$. The hub starting vertex has weight $C^{-1} = o(q/C)$. This proves (11) and controls the route from establishment to fixation.

For a Bd mutant leaf, hub activation has order-one rate and leaf loss has rate $O(C^{-1})$. During an activated excursion, whose return rate is $m + O(1)$, a successful ordinary-core mark occurs with probability $\Theta(1/m)$. Resolving successive excursions gives loss-before-mark probability $O(m/C) + o(1)$; failed finite core families have a geometrically summable conditioned progeny tail. This proves (12). Under dB, the leaf dies at rate one after the common death-clock time change, whereas hub activation is at most r/C , proving (13).

At reciprocal fitness $s = 1/r$, the ordinary core family before its first hub change is uniformly subcritical. Its killed Green function has bounded expected total particle-time and an exponentially decreasing occupation tail, uniformly when s ranges over the reciprocal compact. Since the hub-change hazard is $O(i/C)$ at ordinary count i , an ordinary start changes the hub with probability $O(C^{-1})$, and the ordinary count at that event is tight.

We need the sharper fact that this rare event is not itself fixation. Under Bd, a mutant hub reverts at rate $m + O(1)$, while its total rate of creating a new core or leaf mutant is $O(1 + m/C)$; its probability of a productive excursion is therefore $O(m^{-1}) + O(C^{-1}) = o(1)$. Under dB, the mutant-hub lifetime is tight. It may create $O(m)$ mutant leaves, but after hub loss their total chance of reactivating it before they die is $O(m/C) = o(1)$, and the number of core descendants remains tight. Exponentially long hub episodes have probability $o(1)$.

For completeness, a two-pass renewal argument makes the little-oh gain explicit. For fixed L , let $x_{C,L}^U$ be the largest fixation probability among states with resident hub, no mutant pendants, and at most L mutant ordinary vertices; define $y_{C,L}^U$ analogously with a mutant hub. The killed-Green bounds above first give the coarse estimate

$$x_{C,L}^U \leq e^{-\kappa C} + A_L/C = O_L(C^{-1}).$$

Indeed, before the hub changes such a state must either create a macroscopic core family, at exponentially small cost, or incur a hub-change event, whose integrated hazard is at most A_L/C . On a mutant-hub excursion, the Bd and dB estimates in the preceding paragraph show that direct productive escape has probability $e_C = o(1)$. After an unproductive return, the hub is resident; after all surviving pendants die without reactivation, the number of ordinary descendants is tight. Thus for every fixed L there are tails $\tau_{L,L'} \downarrow 0$ as $L' \rightarrow \infty$, uniformly in C , such that

$$y_{C,L}^U \leq e_C + x_{C,L'}^U + \tau_{L,L'}.$$

For dB, after truncating the ordinary cloud at L' , its contribution to hub reactivation while the leaves clear is $O_{L'}(C^{-1})$; the complementary cloud event is included in $\tau_{L,L'}$. Taking first $C \rightarrow \infty$ and then $L' \rightarrow \infty$ proves $y_{C,L}^U = o(1)$. Finally return to a single ordinary mutant. Conditional on its first hub change, the ordinary count has an exponentially decreasing tail. Truncating that count at L and using the refined hub-excursion bound gives

$$x_{C,1}^U \leq e^{-\kappa C} + \frac{A}{C}(y_{C,L}^U + \tau_L), \quad \tau_L \downarrow 0.$$

Hence $x_{C,1}^U = o(C^{-1})$, first letting $C \rightarrow \infty$ and then $L \rightarrow \infty$, while a hub start has fixation probability $o(1)$. Averaging over clique starts, or weighting them by reciprocal degree for J_H , now gives $o(C^{-1})$ in both cases. The initialization identities (14)–(15) now follow by averaging the $C - 1$ ordinary clique starts, the hub start, and the m pendant starts. The exceptional hub contribution has normalized mass $O(C^{-1}) = o(q/C)$. $\square$

6 Pair gates and the complete post-establishment path

For the center, let

$$I_H = \sum_{x \text{ in clique}} \frac{1}{d_x} = 1 + \frac{1}{C-1+m}, \quad J_H(r) = \sum_{x \text{ in clique}} \frac{h_H^{\text{dB}}(\{x\}; r)}{d_x},$$

where h_H^{dB} is the isolated-center dB fixation committor. For a pair with internal weight $W = C/\sigma$, put

$$I_P = \frac{2}{W} = \frac{2\sigma}{C}, \quad J_P(r) = \frac{1}{W} = \frac{\sigma}{C}.$$

The second identity uses the exact dB singleton value $1/2$ on K_2 . The center estimates imply

$$I_H = 1 + o(1), \quad J_H(r) = p + o(1),$$

while $J_H(1/r)$ and the reciprocal core fixation probability are $o(C^{-1})$.

Proposition 6 (Gate odds and global sweep). *For every fixed $r > 1$, the separated macro-fixation probabilities satisfy*

$$P_U^H = 1 - o(q/C), \quad P_{\text{Bd}}^P \longrightarrow \frac{Z_B}{1 + Z_B}, \quad P_{\text{dB}}^P \longrightarrow \frac{Z_D}{1 + Z_D},$$

where

$$Z_B = \sigma(r^2 - 1), \quad Z_D = \frac{2r(r-1)}{\sigma}.$$

The convergence is uniform when r ranges over a compact subinterval of $(1, \infty)$.

Proof. For one specified pair, let A, D, B, C' denote, respectively, mutant-pair invasion of a resident center, resident-center recovery of that pair, mutant-center invasion of a resident pair, and resident-pair recovery of the center. In each row below a_P means the isolated pair fixation a_P^U for that update rule. Summing the parent-target or death-parent events gives, after a common time change,

	A	D	B	C'
Bd	$CrI_P u_{\text{core}}^{\text{Bd}}(r)$	$2I_H a_P(1/r)$	$2rI_H a_P(r)$	$CI_P u_{\text{core}}^{\text{Bd}}(1/r)$
dB	$2rJ_H(r)$	$(C/r)J_P(1/r)$	$CrJ_P(r)$	$(2/r)J_H(1/r)$

The isolated pair values are

$$a_P^{\text{Bd}}(r) = \frac{r}{r+1}, \quad a_P^{\text{Bd}}(1/r) = \frac{1}{r+1}, \quad a_P^{\text{dB}}(r) = a_P^{\text{dB}}(1/r) = \frac{1}{2}.$$

On macro states the only nonzero rates are

$$\begin{aligned} (0, k) &\rightarrow (1, k) : kA, & (0, k) &\rightarrow (0, k-1) : kD, \\ (1, k) &\rightarrow (1, k+1) : (q-k)B, & (1, k) &\rightarrow (0, k) : (q-k)C'. \end{aligned}$$

From $(1, k)$, the chance that the center reverses before the next pair conversion is $C'/(B + C')$. A union bound over at most q conversions gives

$$1 - P_U^H \leq q \frac{C'}{B + C'}.$$

Proposition 5 and the portal estimates give $qC'/B = o(q/C)$, uniformly on each fitness compact. Thus a mutant center converts all resident pairs with probability $1 - o(q/C)$.

From $(0, 1)$, the first successful macro event is either center invasion at rate A or loss of the sole mutant pair at rate D . Hence

$$P_U^P = \frac{A}{A+D} p_{1,1},$$

where $p_{1,1}$ is the macro-fixation probability from $(1, 1)$. The same sweep bound, now with at most $q - 1$ remaining conversions, gives $p_{1,1} \rightarrow 1$. (Equivalently, $P_U^P = [A/(A+D)][(B+C')/B]P_U^H$.) The rate table and the isolated K_2 values give

$$Z_B = \sigma(r^2 - 1), \quad Z_D = \frac{2r(r-1)}{\sigma}.$$

More precisely, $A/D \rightarrow Z_U$, which yields the stated limits. The argument uses the exact absorbing macro chain, including adverse reversals; no branching survival probability is substituted for global fixation. $\square$

7 The simultaneous correction and its optimization

Put $\eta = q/C$. Substitution of (14)–(15) and Proposition 6 into the finite trace gives the key asymptotic statement.

Proposition 7 (Two response functions). *For $C = t^4$, $q = t$, $m/q \rightarrow \lambda$, and fixed $\sigma > 0$, the separated trace satisfies*

$$\frac{\rho_{\text{Bd},t}^0(r)}{\rho_{\text{Bd}}(K_{n_t}, r)} = 1 + \eta \left[\frac{2(\sigma - 1)}{1 + \sigma(r^2 - 1)} + \frac{\lambda}{r - 1} \right] + o_r(\eta), \quad (21)$$

$$\frac{\rho_{\text{dB},t}^0(r)}{\rho_{\text{dB}}(K_{n_t}, r)} = 1 + \eta \left[\frac{2\{r(2 - r) - \sigma\}}{\sigma + 2r(r - 1)} - \lambda \right] + o_r(\eta). \quad (22)$$

The remainders are uniform on every compact subinterval of $(1, \infty)$.

Proof. The complete-graph Bd baseline differs from p exponentially in C , while the dB baseline differs from p by $O(C^{-1})$; both errors are $o(\eta)$. The center terms follow from (14)–(15). The pair terms follow by the exact simplifications

$$2 \left\{ \frac{[r/(r+1)]Z_B/(1+Z_B)}{p} - 1 \right\} = \frac{2(\sigma - 1)}{1 + \sigma(r^2 - 1)}, \quad (23)$$

$$2 \left\{ \frac{[1/2]Z_D/(1+Z_D)}{p} - 1 \right\} = \frac{2\{r(2 - r) - \sigma\}}{\sigma + 2r(r - 1)}. \quad (24)$$

One leaf replaces one core vertex and contributes $1/p - 1 = 1/(r - 1)$ under Bd and -1 under dB. All inputs are compact-uniform by Propositions 4, 5, and 6, so the combined remainders are as stated. $\square$

Lemma 8 (Exact tangency). *At (σ_*, λ_*) , both response functions in (21)–(22) are strictly positive for every $1 < r < R_{\text{hyb}}$ and vanish simultaneously at $r = R_{\text{hyb}}$.*

Proof. Both bracketed corrections are positive precisely when

$$L(r, \sigma) < \lambda < U(r, \sigma),$$

where

$$L = \frac{2(1 - \sigma)(r - 1)}{1 + \sigma(r^2 - 1)}, \quad U = \frac{2\{r(2 - r) - \sigma\}}{\sigma + 2r(r - 1)}.$$

After multiplication by positive denominators, $L < U$ is equivalent to

$$F_r(\sigma) < 0,$$

where

$$F_r(\sigma) = (r - 1)\sigma^2 + (r^3 - 4r^2 + 3r + 1)\sigma + r(2r - 3).$$

Its minimizing value and minimum are

$$\sigma(r) = \frac{-r^3 + 4r^2 - 3r - 1}{2(r - 1)}, \quad \min_{\sigma} F_r(\sigma) = -\frac{P(r)}{4(r - 1)}.$$

Exact Sturm counting gives no root of P in $(1, 3/2)$ and exactly one in $(3/2, 151/100)$. At that root, σ_* is the double tangency and

$$L(R_{\text{hyb}}, \sigma_*) = U(R_{\text{hyb}}, \sigma_*) = \lambda_*.$$

For fixed σ_* , L is increasing and U decreasing on $(1, R_{\text{hyb}})$ because

$$\partial_r L = \frac{2(\sigma - 1)\{\sigma(r - 1)^2 - 1\}}{\{1 + \sigma(r^2 - 1)\}^2} > 0, \quad \partial_r U = \frac{4r(\sigma - r)}{\{2r(r - 1) + \sigma\}^2} < 0.$$

Here $0 < \sigma_* < 1 < r$ and $\sigma_*(r - 1)^2 < 1$ throughout the isolating interval, so the displayed factorizations determine the signs without numerical approximation. Thus

$$L(r, \sigma_*) < \lambda_* < U(r, \sigma_*) \quad (1 < r < R_{\text{hyb}}),$$

which proves the claim. $\square$

Proof of Theorem 2. Fix $r \in (1, R_{\text{hyb}})$. Then $r \in I_t$ for every sufficiently large t . By Proposition 7 and Lemma 8, the two separated normalized gains, after multiplication by n_t/q_t , converge to strictly positive numbers. The defining inequality for e_t changes each scaled gain by at most $1/t$. Thus both actual connected-graph gains are strictly positive for all sufficiently large t . The sequence, including every weak-edge weight, was fixed before r was chosen, which is exactly the quantifier order in the definition of R_{sim} . $\square$

Proposition 9 (Architecture-optimality). *Within the dilute heavy-pair plus hub-leaf leading regime (21)–(22), R_{hyb} is the exact largest threshold of an interval beginning at one on which a fixed pair (σ, λ) can make both corrections positive.*

Proof. For fixed positive σ , feasibility implies $F_r(\sigma) < 0$, and hence $\min_{s \in \mathbb{R}} F_r(s) < 0$. At R_{hyb} the unconstrained minimum is zero at the admissible value $0 < \sigma_* < 1$. Immediately above R_{hyb} the minimum is positive, so no positive σ —indeed no real σ —is feasible. Since the fixed pair (σ_*, λ_*) works throughout $(1, R_{\text{hyb}})$, no interval beginning at one can extend farther. $\square$

Corollary 10 (A rational family). *Taking $\sigma = 19/137$ and $\lambda = 20/27$ gives, exactly at $r = 3/2$,*

$$B = \frac{232}{17361} > 0, \quad D = \frac{65}{12123} > 0.$$

This completely rational parameter choice has threshold

$$R_{\mathbb{Q}} = \frac{5069 + 12\sqrt{147001}}{6439} = 1.50176815223369\dots > 3/2.$$

Thus a family with rational internal ratios and rational positive weak edges already proves a strict interval beyond $3/2$.

Proof. Substitution into (21)–(22) gives the two displayed margins. For these rational parameters the dB response first vanishes at the unique root in $(1, 2)$ of

$$6439r^2 - 10138r + 703 = 0,$$

which is $R_{\mathbb{Q}}$; the Bd response is still strictly positive there. Both responses are therefore positive on $(1, R_{\mathbb{Q}})$ by the same monotonicity calculation used in Lemma 8. The weak edges remain rational as well: repeat the least-dyadic diagonal construction with $I_t = [1 + 1/t, R_{\mathbb{Q}} - 1/t]$, $\sigma = 19/137$, and $m_t = \lfloor (20/27)t \rfloor$. Every edge-weight ratio in this specialized family is therefore rational. $\square$

8 Discussion

The central contribution is the quantifier in Theorem 2. One weighted graph sequence is chosen without knowing fitness and works for each fixed r in an interval extending beyond $3/2$. A family tuned separately at each fitness would establish a weaker statement. The diagonal choice of ε_t is therefore part of the construction, not merely a technical limit: compact convergence selects one positive cut that works simultaneously on an expanding interval of fitnesses.

The mechanism uses two modules with opposite response signatures. Heavy pairs contribute positively to dB and negatively to Bd, whereas hub pendants contribute positively to Bd and negatively to dB. Their densities are vanishing, so the well-mixed core supplies the baseline and the module responses appear at the common scale q_t/C_t . The exact Schur trace is important here. It resolves every local excursion before the next weak-cut event and retains the possibility that a nearly successful global sweep reverses; replacing it by an independent-lineage approximation would not prove fixation.

Biologically, Bd and dB idealize two different ways local turnover can be resolved: reproduction can select a parent before a replacement site, or a vacancy can first determine which neighbors compete. The construction shows that robustness to this order need not require most individuals to occupy a special environment. A vanishing fraction of paired niches and hub-associated peripheral sites can bias invasion in complementary ways while the bulk remains well mixed. The result is qualitative rather than a claim of empirical optimality: it identifies a mechanism by which rare structural heterogeneities can overcome update-order antagonism in this beneficial-fitness regime.

There are two sharp limitations. Proposition 9 optimizes only the displayed pair–pendant leading-order response cone. It does not preclude other modules, non-dilute interactions, or a different population-scale hierarchy from extending the interval. Conversely, the paper supplies no finite universal upper bound on R_{sim} . Establishing such a bound would require a comparison that couples Bd and dB fixation on arbitrary growing weighted graphs, rather than a classification of this construction.

The quantitative advantage is also asymptotically small. Here $q_t/C_t = t^{-3} = \Theta(n_t^{-3/4})$, so the normalized fixation gain tends to zero. The pair weights grow like C_t , while the exact diagonal may select very small positive weak edges. The proof gives eventual positivity for each fixed fitness but no useful finite-size bound on $t_0(r)$, and it makes no claim of empirical or finite-population optimality.

Statements and Declarations

Data and code availability. No empirical data are used. The LaTeX source, exact symbolic and rational verifiers, labelled-chain audit, build scripts, and adversarial review record accompany the manuscript in the development repository <https://github.com/AlecKriebel/Math>. Version 1.0.0 is contained in the full-repository software snapshot archived at <https://doi.org/10.5281/zenodo.21852072>; that unrefereed release contains the v1 manuscript together with its source and software, and is not a peer-reviewed publication. The present manuscript is a major superseding revision, and its standalone deterministic bundle is intended for versioned deposit with the preprint. The replay enumerates all 512 configurations and 108 orbit fibres of its finite audit graph and independently checks the response algebra, rational margins, Sturm isolation, and tangency. These certificates audit finite transition aggregation and exact algebra; the weak-cut and population asymptotics are analytic proofs in the manuscript, not computer-assisted consequences of the replay.

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AI-assisted research disclosure. Generative-AI systems were used substantively in mathematical exploration, derivation, adversarial proof analysis, software development, and manuscript preparation. The author determined the scope and claims and accepts responsibility for the manuscript and accompanying materials. All exact certificates are supplied for independent replay, and numerical evidence is explicitly separated from proof.

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